

## PROBABILITY, STATISTICS AND COMPLEX

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| III Semester: MECH , AERO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                |              |   |   |         |               |     |       |
| Course Code:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Category                                       | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6BS05                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | BSC                                            | L            | T | P | C       | CIA           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                | 3            | 1 | - | 4       | 40            | 60  | 100   |
| <b>COURSE OBJECTIVES</b><br>1.Evaluation of Probability distribution of Discrete and Continuous random<br>2. Evaluation of the given data for appropriate test of hypothesis for large samples.<br>3. Evaluation of the given data for appropriate test of hypothesis for small samples.<br>4. Evaluation of Limit, continuity, differentiability, analyticity and finding harmonic function<br>5. Evaluation of the line integrals along piece wise smooth paths& Express the given complex function as a power series using Taylor's series, Maclaurin's series, Residues and improper integrals.                          |                                                |              |   |   |         |               |     |       |
| UNIT-I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | RANDOM VARIABLES AND PROBABILITY DISTRIBUTIONS |              |   |   |         |               |     |       |
| Random Variables – Discrete and Continuous. Probability distributions, mass function/ density function of a probability distribution, mathematical expectation,.<br>Binomial, Poisson, Normal distributions, and their Properties and hence find mean and variance                                                                                                                                                                                                                                                                                                                                                           |                                                |              |   |   |         |               |     |       |
| UNIT-II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | TESTING OF HYPOTHESIS - I                      |              |   |   |         |               |     |       |
| <b>Testing of hypothesis:</b> Null hypothesis, Alternate hypothesis, Type I& Type II errors – critical region, confidence interval, Level of significance. One sided test, Two sided test.<br><b>Large sample tests:</b><br>(i) Test of Equality of means of two samples, equality of sample mean and population mean (cases of known variance & unknown variance, equal and unequal variances)<br>(ii) Tests of significance of difference between sample S.D and population S.D.<br>(iii) Tests of significance difference between sample proportion and population proportion, difference between two sample proportions. |                                                |              |   |   |         |               |     |       |
| UNIT-III                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | TESTING OF HYPOTHESIS-II                       |              |   |   |         |               |     |       |
| Student t-distribution, its properties; Test of significance sample mean and population mean, difference between means of two small samples. Snedecor's F- distribution and its properties. Test of equality of two population variances. Chi-square distribution, its properties, Chi-square test of goodness of fit.                                                                                                                                                                                                                                                                                                       |                                                |              |   |   |         |               |     |       |
| UNIT-IV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | COMPLEX DIFFERENTIATION                        |              |   |   |         |               |     |       |
| Limit, Continuity and Differentiation of Complex functions, Analyticity, Cauchy-Riemann equations (without proof), finding harmonic conjugate.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                |              |   |   |         |               |     |       |
| UNIT-V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | COMPLEX INTEGRATION                            |              |   |   |         |               |     |       |
| Line integral, Cauchy's theorem, Cauchy's Integral formula, Zeros of analytic functions, Singularities, Taylor's series, Laurent's series, Residues, Cauchy Residue theorem (All theorems without Proof).                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                |              |   |   |         |               |     |       |
| Text Books:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                |              |   |   |         |               |     |       |
| 1. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.<br>2. Probability and Statistics for Engineers by Richard Arnold Johnson, Irwin Miller and John E. Freund, New Delhi, Prentice Hall.                                                                                                                                                                                                                                                                                                                                                                                                   |                                                |              |   |   |         |               |     |       |

3. Probability and Statistics for Engineers and Sciences by Jay L. Devore, Cengage Learning

**Reference Books:**

1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
2. Fundamentals of Mathematical Statistics by S.C. Guptha & V.K. Kapoor, S. Chand
3. Introduction to Probability and Statistics for Engineers and Scientists by Sheldon M. Ross, Academic Press

**COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO**

At the end of the course, student will be able to:

1. Describe the Discontinuous random variables and Find moments of Discrete and continuous random variables.
2. Evaluate the given data for appropriate test of hypothesis for Large samples.
3. Evaluate the given data for appropriate test of hypothesis for small samples.
4. Evaluate Limit, continuity, differentiability, analyticity and harmonic function .
5. Express the given complex function as a power series using Taylor's series and Maclaurin's series and Evaluation of Residues